Operations with radical expressions

1

a $\sqrt{361}$

b $\sqrt{323}$

2

a 45

b 5

c $\sqrt{22}$

d $60\sqrt{13}$

e $4\sqrt{2}$

f $15\sqrt{35}$

g $\sqrt{70}$

h $5\sqrt{7}$

i $20\sqrt{7}$

j $176\sqrt{3}$

k $35\sqrt{21}$

3

a

i 2.6

ii 4.5

iii 5.2

b No

c No

4 False

5

a $26\sqrt{3}$

b $11\sqrt{10}$

c $-15\sqrt{3}$

d $6\sqrt{7}$

e $-15\sqrt{5}$

f $-34\sqrt{10}$

g $34\sqrt{3}$

h $17\sqrt{10}$

i $28\sqrt{2}$

j $-9\sqrt{2}$

k $-19\sqrt{2}$

l $-18\sqrt{5}$

m $-9\sqrt{10}$

n $-21\sqrt{7}$

o $39\sqrt{3}+45\sqrt{11}$

p $17\sqrt{5}-\sqrt{3}$

q $20\sqrt{6}+5\sqrt{11}$

r $28\sqrt{2}+38\sqrt{3}$

s $23\sqrt{13}+10\sqrt{6}$

t $41\sqrt{3}+12\sqrt{7}$

u $4+3\sqrt{5}$

v $\frac{3\sqrt{3}}{2}$

6

a $4\sqrt{6}$

b $10\sqrt{3}$

c $16\sqrt{5}$

d $6\sqrt{6}$

e $-3\sqrt{2}$

f $42\sqrt{5}$

g $-8\sqrt{3}$

7 $7\sqrt{5}\ \text{ft}^2$

8 $3\sqrt{5}$ square units

9 $12\sqrt{2}$ units

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10

a $14\sqrt{3}$ cm

b 36 cm^2

11 $10 + \sqrt{62}$ mi

12 $12\sqrt{5} + 10\sqrt{2}$ ft

13

a $\frac{5\sqrt{129}}{28}$ m^2

b 2.03 m^2

Additional questions

14 $\frac{\sqrt{287}}{10}$ m^2

15 She didn't make an error.

16

a $\sqrt{1374}$ yd

b 26 yd

c $(\sqrt{1374} - 26)$ yd

17 Any two pairs of values where either k or m (or both) are zero. For example:
$k = 0$ $m = 4$
$k = 2$ $m = 0$

18 When adding two radicals you do not combine the radicands. Instead, you can solve by

$$\begin{aligned} 5\sqrt{5} + \sqrt{45} &= 5\sqrt{5} + 3\sqrt{5} \\ &= 8\sqrt{5} \end{aligned}$$

19 Sasha is correct. Her work would have looked like

$$\begin{aligned} \sqrt[3]{54} - \sqrt[3]{128} &= \sqrt[3]{3*3*3*2} - \sqrt[3]{2*2*2*2*2*2*2} \\ &= 3\sqrt[3]{2} - 2*2\sqrt[3]{2} \\ &= 3\sqrt[3]{2} - 4\sqrt[3]{2} = -\sqrt[3]{2} \end{aligned}$$

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